



MERIT EDUCATION GROUP

Perimeter, Area & Algebraic Expressions

I can write and evaluate algebraic expressions for perimeter and area.

Standard: 9.3.5.7

Standard: 9.3.5.8

Day 15 · Unit 1 · Theme: Generic Amusement Park

1 BELL RINGER — WHITEBOARDS READY!

Write an expression for the perimeter of a square with side length s .

Your response: _____

2 KEY VOCABULARY

Perimeter — The _____ around a shape. _____

Area — The amount of _____ inside a shape. _____

Dimension — A measurable side of a shape: length, _____, or height. _____

3 DIRECT INSTRUCTION — TAKE NOTES (I DO)

Example: A rectangle's length is 3 more than its width w . Write an expression for the perimeter.

• Step 1 — Length in terms of w : $l = \text{_____} \cdot (w + 3)$

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- Step 2 — Perimeter formula: $P = 2l + 2w = 2(\text{_____}) + 2w \cdot (w + 3)$
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- Step 3 — Distribute and combine: $2w + 6 + 2w = \text{_____} \cdot (4w + 6)$
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4 GUIDED PRACTICE — WORK TOGETHER (WE DO)

- A square has side s . Write expressions for area and perimeter. A triangle has sides x , $2x$, and $x + 5$. Write its perimeter.
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5 CULTURE CONNECTION — MINNESOTA STORIES

Sami (Sápmi / Northern Europe) — The Sámi are the Indigenous people of Sápmi, spanning northern Norway, Sweden, Finland, and Russia. Sami culture centers on reindeer herding and duodji (handicraft).

Word problem: A Sámi garment maker decorates a rectangular luhkka (reindeer herder) cloak. The length is $3x + 2$ and the width is $2x - 1$, all in units. Write an expression for the perimeter of the rectangle.

6 INDEPENDENT PRACTICE — YOU TRY!

- A rectangle has area 48 and width 6. Write and solve an equation for the length l .
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A rectangle has length $(x + 4)$ and width 3. Write an expression for its perimeter.

Answer: _____

Show your work for full credit.